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United States Patent	12397792
Kind Code	B2
Date of Patent	August 26, 2025
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Vehicle driving assist apparatus, vehicle driving assist method, and vehicle driving assist program

Abstract

A vehicle driving assist apparatus executes a constant speed moving control to autonomously control a moving speed of an own vehicle to a set moving speed. The vehicle driving assist apparatus pauses the constant speed moving control and executes an intermittent braking control when the own vehicle moves down a slope having a gradient smaller than a predetermined gradient threshold while the constant speed moving control is executed. The intermittent braking control is a control to alternately apply and stop applying the braking force to the own vehicle. The braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control.

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Appl. No.:	17/858599
Filed:	July 06, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230082526 A1	Mar. 16, 2023

Foreign Application Priority Data

JP	2021-149230	Sep. 14, 2021
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Publication Classification

Int. Cl.: B60W30/18 (20120101)

U.S. Cl.:

CPC B60W30/18109 (20130101); B60W2510/18 (20130101); B60W2552/15 (20200201)

Field of Classification Search

CPC: B60W (30/18109); B60W (2552/15); B60W (2510/18)

USPC: 701/10

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese patent application No. JP 2021-149230 filed on Sep. 14, 2021, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND

Field

(2) The invention relates to a vehicle driving assist apparatus, a vehicle driving assist method, and a vehicle driving assist program.

Description of the Related Art

(3) There is known a vehicle driving assist apparatus which executes a constant speed moving control to maintain a moving speed of an own vehicle at a set moving speed by autonomously controlling a driving force and a braking force applied to the own vehicle without a driver of the own vehicle operating an accelerator pedal and a brake pedal (for example, see JP 2018-31467 A).

(4) The known vehicle driving assist described above may continue applying the braking force to the own vehicle to maintain the moving speed of the own vehicle at the set moving speed when the own vehicle moves down a slope. If the slope where the own vehicle move down is moderate, the known vehicle driving assist apparatus may continue applying the relatively small braking force to the own vehicle for a relatively long time.

(5) If the small braking force has been applied to the own vehicle for a long time, there may occur a vibration such as a so-called self-excited vibration due to thermal expansion of brake pads and brake discs. Such a vibration may lead to a discomfort of the driver of the own vehicle.

SUMMARY

(6) An object of the invention is to provide a vehicle driving assist apparatus, a vehicle driving assist method, and a vehicle driving assist program which can prevent an occurrence of the self-excited vibration due to the constant speed moving control.

(7) According to the invention, a vehicle driving assist apparatus comprises an electronic control unit which executes a constant speed moving control to autonomously control a moving speed of an own vehicle to a set moving speed. The electronic control unit is configured to pause the constant speed moving control and execute an intermittent braking control when the own vehicle moves down a slope having a gradient smaller than a predetermined gradient threshold while the constant speed moving control is executed. The intermittent braking control is a control to alternately apply and stop applying the braking force to the own vehicle. The braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control.

(8) As described above, when the own vehicle moves down the moderate slope while the constant speed moving control is executed, the relatively small braking force may have been applied to the own vehicle for a long time. When the small braking force has been applied for a long time, there may occur a vibration such as a so-called self-excited vibration due to thermal expansion of the brake pads and the brake discs. Such a vibration may lead to a discomfort of the driver of the own vehicle.

(9) With the vehicle driving assist apparatus according to the invention, on the condition that the own vehicle moves down the slope having the gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed, the intermittent braking control is executed. The braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control. Thus, the occurrence of the self-excited vibration can be prevented.

(10) According to an aspect of the invention, the electronic control unit may be configured to determine that the own vehicle moves down the slope having the gradient smaller than the predetermined gradient threshold when the braking force applied to the own vehicle by the constant speed moving control has been smaller than a predetermined braking force threshold for a

predetermined time.

(11) With the vehicle driving assist apparatus according to this aspect of the invention, whether the own vehicle moves down the moderate slope can be determined, based on the braking force applied to the own vehicle.

(12) According to another aspect of the invention, the electronic control unit may be configured to pause the intermittent braking control and start the constant speed moving control when the moving speed of the own vehicle becomes equal to or greater than a predetermined upper limit vehicle moving speed greater than the set moving speed, or becomes equal to or smaller than a predetermined lower limit vehicle moving speed smaller than the set moving speed while the intermittent braking control is executed.

(13) When the intermittent braking control is stopped, and the constant speed moving control is started in response to the moving speed of the own vehicle considerably increasing while the intermittent braking control is executed, the relatively great braking force is applied to the own vehicle by the constant speed moving control. Thus, even if the constant speed moving control is started, the self-vibration may not occur. Thus, in order to meet a request of the driver to move the own vehicle at a constant speed, the intermittent braking control should be stopped, and the constant speed moving control should be started.

(14) Further, when the intermittent braking control is stopped, and the constant speed moving control is started in response to the moving speed of the own vehicle considerably decreasing while the intermittent braking control is executed, the considerably small braking force is probably applied to the own vehicle by the constant speed moving control. Thus, even if the constant speed moving control is started, the self-vibration may not occur. Thus, in order to meet a request of the driver to move the own vehicle at a constant speed, the intermittent braking control should be stopped, and the constant speed moving control should be started.

(15) With the vehicle driving assist apparatus according to this aspect of the invention, on the condition that the moving speed of the own vehicle considerably increases or decreases while the intermittent braking control is executed, the intermittent braking control is stopped, and the constant speed moving control is started. Thus, the request of the driver can be met.

(16) Further, according to the invention, a vehicle comprises the vehicle driving assist apparatus according to the invention.

(17) Furthermore, a vehicle driving assist method according to the invention is a method of executing a constant speed moving control to autonomously control a moving speed of an own vehicle to a set moving speed. The vehicle driving assist method according to the invention comprises a step to pause the constant speed moving control and execute an intermittent braking control when the own vehicle moves down a slope having a gradient smaller than a predetermined gradient threshold while the constant speed moving control is executed. The intermittent braking control is a control to alternately apply and stop applying the braking force to the own vehicle. The braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control.

(18) A vehicle driving assist program according to the invention is a program of executing a constant speed moving control to autonomously control a moving speed of an own vehicle to a set moving speed. The vehicle driving assist program according to the invention is configured to pause the constant speed moving control and execute an intermittent braking control when the own vehicle moves down a slope having a gradient smaller than a predetermined gradient threshold while the constant speed moving control is executed. The intermittent braking control is a control to alternately apply and stop applying the braking force to the own vehicle. The braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control.

(19) Elements of the invention are not limited to elements of embodiments and modified examples of the invention described with reference to the drawings. The other objects, features and

accompanied advantages of the invention can be easily understood from the embodiments and the modified examples of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a view which shows a vehicle driving assist apparatus according to an embodiment of the invention and a vehicle or an own vehicle on which the vehicle driving assist apparatus is installed.
- (2) FIG. 2A is a view which shows a scene that the own vehicle is moved, following a preceding vehicle by a following moving control.
- (3) FIG. 2B is a view which shows a scene that an inter-vehicle distance is longer than a target inter-vehicle distance.
- (4) FIG. 2C is a view which shows a scene that the inter-vehicle distance is shorter than a target inter-vehicle distance.
- (5) FIG. 3 is a view which shows a scene that there is no preceding vehicle.
- (6) FIG. 4 is a view which shows a time chart when an intermittent braking control is executed as a vehicle moving speed increases.
- (7) FIG. 5 is a view which shows a time chart when the intermittent braking control is executed as the vehicle moving speed decreases.
- (8) FIG. 6 is a view which shows a time chart when the intermittent braking control is executed.
- (9) FIG. 7 is a view which shows a flowchart of a routine executed by the vehicle driving assist apparatus according to the embodiment of the invention.
- (10) FIG. 8 is a view which shows a flowchart of a routine executed by the vehicle driving assist apparatus according to the embodiment of the invention.
- (11) FIG. 9 is a view which shows a flowchart of a routine executed by the vehicle driving assist apparatus according to the embodiment of the invention.
- (12) FIG. 10 is a view which shows a flowchart of a routine executed by the vehicle driving assist apparatus according to the embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

- (13) Below, a vehicle driving assist apparatus **10** according to an embodiment of the invention will be described with reference to the drawings. As shown in FIG. 1, the vehicle driving assist apparatus **10** is installed on an own vehicle **100**.
- (14) The vehicle driving assist apparatus **10** includes an ECU **90** as a control unit. The ECU **90** includes a micro-computer as a main component. The ECU **90** includes a CPU, a ROM, a RAM, a non-volatile memory, and an interface. The CPU is configured or programmed to realize various functions by executing instructions, programs, and routines stored in the ROM.
- (15) <Vehicle Moving Apparatus>
- (16) Further, a vehicle moving apparatus **20** is installed on the own vehicle **100**. The vehicle moving apparatus **20** drives, brakes, and steers the own vehicle **100**. In this embodiment, the vehicle moving apparatus **20** includes a driving apparatus **21**, a braking apparatus **22**, and a steering apparatus **23**.
- (17) <Driving Apparatus>
- (18) The driving apparatus **21** outputs a driving force to be applied to the own vehicle **100** to move the own vehicle **100**. In this embodiment, the driving apparatus **21** includes an internal combustion engine. In this regard, the driving apparatus **21** may include at least one electric motor or a combination of the internal combustion engine and the at least one electric motor. The driving apparatus **21** is electrically connected to the ECU **90**. The ECU **90** controls the driving force output from the driving apparatus **21** by controlling operations of the driving apparatus **21**.

(19) <Braking Apparatus>

(20) The braking apparatus **22** outputs a braking force to be applied to the own vehicle **100** to brake the own vehicle **100**. In this embodiment, the braking apparatus **22** includes a hydraulic brake apparatus which applies the braking force to the own vehicle **100** by pressing brake pads to brake discs provided on wheels of the own vehicle **100**, respectively. The braking apparatus **22** is electrically connected to the ECU **90**. The ECU **90** controls the braking force output from the braking apparatus **22** by controlling operations of the braking apparatus **22**.

(21) <Steering Apparatus>

(22) The steering apparatus **23** outputs a steering force to be applied to the own vehicle **100** to steer the own vehicle **100**. The steering apparatus **23** may include a power steering apparatus. The steering apparatus **23** is electrically connected to the ECU **90**. The ECU **90** controls the steering force output from the steering apparatus **23** by controlling operations of the steering apparatus **23**.

(23) <Sensors, Etc.>

(24) Further, an accelerator pedal **41**, an accelerator pedal operation amount sensor **42**, a brake pedal **43**, a brake pedal operation amount sensor **44**, a steering wheel **45**, a steering shaft **46**, a steering angle sensor **47**, a steering torque sensor **48**, a following moving operation device **49**, a vehicle moving speed detecting device **50**, and a surrounding information detecting apparatus **60** are installed on the own vehicle **100**.

(25) <Accelerator Pedal Operation Amount Sensor>

(26) The accelerator pedal operation amount sensor **42** detects an operation amount of the accelerator pedal **41**. The accelerator pedal operation amount sensor **42** is electrically connected to the ECU **90**. The accelerator pedal operation amount sensor **42** sends information on the detected operation amount of the accelerator pedal **41** to the ECU **90**. The ECU **90** acquires the operation amount of the accelerator pedal **41** as an accelerator pedal operation amount AP, based on the information sent from the accelerator pedal operation amount sensor **42**.

(27) The ECU **90** executes an ordinary accelerating-and-decelerating control to (i) calculate and acquire a required driving force or a required driving torque, based on the accelerator pedal operation amount AP and a moving speed of the own vehicle **100**, i.e., an own vehicle moving speed and (ii) control the operations of the driving apparatus **21** so as to output the driving force corresponding to the required driving force. It should be noted that while the ECU **90** executes a following moving control, a constant speed moving control, or an intermittent braking control described later in detail, the ECU **90** determines the driving force necessary to move the own vehicle **100** so as to realize a target acceleration GA_TGT or a target deceleration GD_TGT set by the following moving control, the constant speed moving control, or the intermittent braking control and controls the operations of the driving apparatus **21** so as to output the driving force corresponding to the determined driving force.

(28) <Brake Pedal Operation Amount Sensor>

(29) The brake pedal operation amount sensor **44** detects an operation amount of the brake pedal **43**. The brake pedal operation amount sensor **44** is electrically connected to the ECU **90**. The brake pedal operation amount sensor **44** sends information on the detected operation amount of the brake pedal **43** to the ECU **90**. The ECU **90** acquires the operation amount of the brake pedal **43** as a brake pedal operation amount BP, based on the information sent from the brake pedal operation amount sensor **44**.

(30) The ECU **90** executes the ordinary accelerating-and-decelerating control to (i) calculate and acquire a required braking force or a required braking torque, based on the brake pedal operation amount BP and (ii) control the operations of the braking apparatus **22** so as to output the braking force corresponding to the required braking force. It should be noted that while the ECU **90** executes the following moving control, the constant speed moving control, or the intermittent braking control described later in detail, the ECU **90** determines the braking force necessary to brake the own vehicle **100** so as to realize the target acceleration GA_TGT or the target

deceleration GD_TGT set by the following moving control, the constant speed moving control, or the intermittent braking control and controls the operations of the braking apparatus **22** so as to output the braking force corresponding to the determined braking force.

(31) <Steering Angle Sensor>

(32) The steering angle sensor **47** detects a rotation angle of the steering shaft **46** with respect to a neutral position thereof. The steering angle sensor **47** is electrically connected to the ECU **90**. The steering angle sensor **47** sends information on the detected rotation angle of the steering shaft **46** to the ECU **90**. The ECU **90** acquires the rotation angle of the steering shaft **46** as a steering angle θ , based on the information sent from the steering angle sensor **47**.

(33) <Steering Torque Sensor>

(34) The steering torque sensor **48** detects a torque which a driver DV of the own vehicle **100** inputs to the steering shaft **46** via the steering wheel **45**. The steering torque sensor **48** is electrically connected to the ECU **90**. The steering torque sensor **48** sends information on the detected torque to the ECU **90**. The ECU **90** acquires the torque which the driver DV inputs to the steering shaft **46** via the steering wheel **45** as a driver input torque, based on the information sent from the steering torque sensor **48**.

(35) The ECU **90** acquires a required steering force or a required steering torque, based on the steering angle θ , the driver input torque, and the moving speed of the own vehicle **100**, i.e., the own vehicle moving speed and controls the operations of the steering apparatus **23** so as to output the steering torque corresponding to the required steering torque.

(36) <Following Moving Operation Device>

(37) The following moving operation device **49** is operated by the driver DV of the own vehicle **100**. The following moving operation device **49** includes switches and/or buttons. The switches and/or the buttons may be provided on the steering wheel **45** of the own vehicle **100** or a lever provided on a steering column of the own vehicle **100**.

(38) In this embodiment, the following moving operation device **49** includes a following moving selecting switch, a vehicle moving speed setting switch, a vehicle moving speed increasing button, a vehicle moving speed decreasing button, and an inter-vehicle distance setting button. The following moving operation device **49** is electrically connected to the ECU **90**.

(39) When a driving assist control including (i) the following moving control, (ii) the constant speed moving control, and (iii) the intermittent braking control described later in detail is not executed, and the following moving selecting switch is operated by the driver DV, a predetermined signal is sent to the ECU **90** from the following moving operation device **49**. When the ECU **90** receives the predetermined signal in question, the ECU **90** determines that the driver DV requests executing the driving assist control. When the ECU **90** determines that the driver DV requests executing the driving assist control, and there is a preceding vehicle, the ECU **90** executes the following moving control. On the other hand, when the ECU **90** determines that the driver DV requests executing the driving assist control, and there is no preceding vehicle, the ECU **90** executes the constant speed moving control.

(40) On the other hand, when the driving assist control is executed, and the following moving selecting switch is operated by the driver DV, a predetermined signal is sent to the ECU **90** from the following moving operation device **49**. When the ECU **90** receives the predetermined signal in question, the ECU **90** determines that the driver DV requests terminating the driving assist control. When the ECU **90** determines that the driver DV requests terminating the driving assist control, the ECU **90** terminates the following moving control, the constant speed moving control, or the intermittent braking control. It should be noted that as described later, while the intermittent braking control is executed, the constant speed moving control has been paused. In this case, when the ECU **90** terminates the intermittent braking control, the ECU **90** terminates the paused constant speed moving control.

(41) Further, when the driving assist control is executed, and the vehicle moving speed setting

switch is operated by the driver DV, a predetermined signal is sent to the ECU **90** from the following moving operation device **49**. When the ECU **90** receives the predetermined signal in question, the ECU **90** sets the current moving speed of the own vehicle **100**, i.e., the current own vehicle moving speed as a set vehicle moving speed V_SET used in the constant speed moving control.

(42) Further, when the driving assist control is executed, and the vehicle moving speed increasing button is operated by the driver DV, a predetermined signal is sent to the ECU **90** from the following moving operation device **49**. When the ECU **90** receives the predetermined signal in question, the ECU **90** increases the set vehicle moving speed V_SET . On the other hand, when the driving assist control is executed, and the vehicle moving speed decreasing button is operated by the driver DV, a predetermined signal is sent to the ECU **90** from the following moving operation device **49**. When the ECU **90** receives the predetermined signal in question, the ECU **90** decreases the set vehicle moving speed V_SET .

(43) Further, when the driving assist control is executed, and the inter-vehicle distance setting button is operated by the driver DV, a predetermined signal is sent to the ECU **90** from the following moving operation device **49**. The predetermined signal in question is a requested inter-vehicle distance signal which represents an inter-vehicle distance D requested as a target inter-vehicle distance D_TGT or a predetermined inter-vehicle distance by the driver DV operating the inter-vehicle distance setting button.

(44) As shown in FIG. 2A, the inter-vehicle distance D is a distance between the preceding vehicle **200** and the own vehicle **100**, and the target inter-vehicle distance D_TGT is the inter-vehicle distance D which the following moving control targets. In this embodiment, the preceding vehicle **200** is a vehicle which is within a predetermined distance range ahead of the own vehicle **100** and moves in a traffic lane in which the own vehicle **100** moves, i.e., moves in an own vehicle moving lane $LN1$. It should be noted that in the drawings, a symbol $LN2$ denotes a traffic lane next to the own vehicle moving lane $LN1$, and in this embodiment, denotes a traffic lane in which an oncoming vehicle moves.

(45) In this embodiment, the inter-vehicle distance D which the driver DV can request as the target inter-vehicle distance D_TGT by operating the inter-vehicle distance setting button is one of three distances, i.e., a long distance, a middle distance, and a short distance.

(46) When the ECU **90** receives the requested inter-vehicle distance signal, the ECU **90** sets the target inter-vehicle distance D_TGT , depending on the inter-vehicle distance D represented by the received requested inter-vehicle distance signal, i.e., depending on a requested inter-vehicle distance D_REQ .

(47) In this embodiment, the ECU **90** sets the target inter-vehicle distance D_TGT , depending on the requested inter-vehicle distance D_REQ in consideration of the current own vehicle moving speed $V100$. In this regard, the ECU **90** may set the target inter-vehicle distance D_TGT , depending on the requested inter-vehicle distance D_REQ without considering the current own vehicle moving speed $V100$.

(48) In particular, the ECU **90** sets as the target inter-vehicle distance D_TGT , the inter-vehicle distance D which makes a predicted reaching time TTC correspond to a predetermined time or a predetermined predicted reaching time TTC_REF . The predicted reaching time TTC is time acquired by dividing the inter-vehicle distance D by the current own vehicle moving speed $V100$. In other words, the ECU **90** sets as the target inter-vehicle distance D_TGT , the inter-vehicle distance D which makes a relationship between the current own vehicle moving speed $V100$ and the predetermined predicted reaching time TTC_REF correspond to a relationship of an expression 1 below.

$$TTC_REF = D / V100 \quad (1)$$

(49) When the requested inter-vehicle distance D_REQ is the long distance, the predetermined predicted reaching time TTC_REF is a long time $TTClong$. When the requested inter-vehicle

distance D_REQ is the middle distance, the predetermined predicted reaching time TTC_REF is a middle time TTCmid. When the requested inter-vehicle distance D_REQ is the short distance, the predetermined predicted reaching time TTC_REF is a short time TTCshort.

(50) <Vehicle Moving Speed Detecting Device>

(51) The vehicle moving speed detecting device **50** detects the moving speed of the own vehicle **100**. The vehicle moving speed detecting device **50** may include vehicle wheel rotation speed sensors. The vehicle moving speed detecting device **50** is electrically connected to the ECU **90**. The vehicle moving speed detecting device **50** sends information on the detected moving speed of the own vehicle **100** to the ECU **90**. The ECU **90** acquires the moving speed of the own vehicle **100** as the own vehicle moving speed V100, based on the information sent from the vehicle moving speed detecting device **50**.

(52) <Surrounding Information Detecting Apparatus>

(53) The surrounding information detecting apparatus **60** includes radio wave sensors **61** and image sensors **62**.

(54) <Radio Wave Sensors>

(55) The radio wave sensor **61** detects information on objects around the own vehicle **100** by using radio waves. The radio wave sensor **61** may be a radar sensor such as a millimeter wave radar, a sonic wave sensor such as an ultrasonic wave sensor such as a clearance sonar, or an optical sensor such as a laser radar such as a LiDAR. The radio wave sensors **61** are electrically connected to the ECU **90**. The radio wave sensor **61** transmits radio waves and receives the radio waves reflected on objects, i.e., receives the reflected radio waves. The radio wave sensors **61** send information on the transmitted radio waves and the received reflected radio waves to the ECU **90**. In other words, the radio wave sensors **61** detect objects around the own vehicle **100** and send information on the detected objects to the ECU **90**. The ECU **90** acquires surrounding information IS on the objects around the own vehicle **100**, based on the information such as radio wave information IR or radio wave data sent from the radio wave sensors **61**. The objects detected by using the radio wave sensors **61** may include vehicles, walls, bicycles, and persons.

(56) <Image Sensors>

(57) The image sensor **62** takes images of a view around the own vehicle **100**. The image sensor **62** may be a camera. The image sensors **62** are electrically connected to the ECU **90**. The image sensor **62** takes images of a view around the own vehicle **100** and sends information on the taken images to the ECU **90**. The ECU **90** acquires the surrounding information IS on a situation around the own vehicle **100**, based on the information such as image information IC or image data sent from the image sensors **62**.

(58) <Summary of Operations of Vehicle Driving Assist Apparatus>

(59) Next, operations of the vehicle driving assist apparatus **10** will be described. When the vehicle driving assist apparatus **10** determines that the driver DV requests executing the following moving control, the vehicle driving assist apparatus **10** executes the following moving control or the constant speed moving control as described below.

(60) <Following Moving Control>

(61) As shown in FIG. 2A, when there is the preceding vehicle **200**, the vehicle driving assist apparatus **10** executes the following moving control. The following moving control is a control to maintain the distance between the own vehicle **100** and the preceding vehicle **200** at a predetermined distance by autonomously controlling the operations of the driving apparatus **21** and/or the braking apparatus **22**. That is, the following moving control is a control to maintain the distance between the own vehicle **100** and the preceding vehicle **200** at the predetermined distance by controlling the operations of the driving apparatus **21** and/or the braking apparatus **22** even when the driver DV does not operate the accelerator pedal **41** or the brake pedal **43**. It should be noted that the vehicle driving assist apparatus **10** determines whether there is the preceding vehicle **200**, based on the surrounding information IS.

(62) When the vehicle driving assist apparatus **10** starts the following moving control, the vehicle driving assist apparatus **10** acquires a difference between the inter-vehicle distance D and the target inter-vehicle distance D_TGT as an inter-vehicle distance difference dD and maintains the inter-vehicle distance D at the target inter-vehicle distance D_TGT by a feedback control such as a PID control to control the operations of the driving apparatus **21** and/or the braking apparatus **22** so as to control the inter-vehicle distance difference dD to zero.

(63) In particular, as shown in FIG. 2B, when the inter-vehicle distance D is greater than the target inter-vehicle distance D_TGT , the vehicle driving assist apparatus **10** calculates the target acceleration GA_TGT for decreasing the inter-vehicle distance D and controls the operations of the driving apparatus **21** so as to realize the calculated target acceleration GA_TGT . Thereby, the own vehicle moving speed $V100$ is increased and as a result, the inter-vehicle distance D is decreased toward the target inter-vehicle distance D_TGT .

(64) On the other hand, as shown in FIG. 2C, when the inter-vehicle distance D is smaller than the target inter-vehicle distance D_TGT , the vehicle driving assist apparatus **10** calculates the target deceleration GD_TGT for increasing the inter-vehicle distance D and controls the operations of the braking apparatus **22** and/or the driving apparatus **21** so as to realize the calculated target deceleration GD_TGT . Thereby, the own vehicle moving speed $V100$ is decreased and as a result, the inter-vehicle distance D is increased toward the target inter-vehicle distance D_TGT .

(65) It should be noted that if the target inter-vehicle distance D_TGT is equal to or smaller than zero, the own vehicle **100** contacts the preceding vehicle **200**. Thus, in this embodiment, the target inter-vehicle distance D_TGT is set to a value greater than zero and enough to decelerate the own vehicle **100** so as not to contact the preceding vehicle **200** if the preceding vehicle **200** decelerates suddenly.

(66) <Constant Speed Moving Control>

(67) As shown in FIG. 3, when there is no preceding vehicle **200**, the vehicle driving assist apparatus **10** executes the constant speed moving control. The constant speed moving control is a control to maintain the own vehicle moving speed $V100$ at the set vehicle moving speed V_SET by autonomously controlling the operations of the driving apparatus **21** and/or the braking apparatus **22**. In other words, the constant speed moving control is a control to maintain the own vehicle moving speed $V100$ at the set vehicle moving speed V_SET by controlling the operations of the driving apparatus **21** and/or the braking apparatus **22** even when the driver DV does not operate the accelerator pedal **41** or the brake pedal **43**.

(68) When the vehicle driving assist apparatus **10** starts the constant speed moving control, the vehicle driving assist apparatus **10** acquires a difference between the own vehicle moving speed $V100$ and the set vehicle moving speed V_SET as a vehicle moving speed difference dV and maintains the own vehicle moving speed $V100$ at the set vehicle moving speed V_SET by a feedback control such as a PID control to control the operations of the driving apparatus **21** and/or the braking apparatus **22** so as to control the vehicle moving speed difference dV to zero.

(69) In particular, when the own vehicle moving speed $V100$ is greater than the set vehicle moving speed V_SET , the vehicle driving assist apparatus **10** calculates the target deceleration GD_TGT for decreasing the own vehicle moving speed $V100$ and controls the operations of the braking apparatus **22** and/or the driving apparatus **21** so as to realize the calculated target deceleration GD_TGT . Thereby, the own vehicle moving speed $V100$ is decreased toward the set vehicle moving speed V_SET .

(70) On the other hand, when the own vehicle moving speed $V100$ is smaller than the set vehicle moving speed V_SET , the vehicle driving assist apparatus **10** calculates the target acceleration GA_TGT for increasing the own vehicle moving speed $V100$ and controls the operations of the driving apparatus **21** so as to realize the calculated target acceleration GA_TGT . Thereby, the own vehicle moving speed $V100$ is increased toward the set vehicle moving speed V_SET .

(71) When the own vehicle **100** moves down a slope or a downward slope while the constant speed

moving control is executed, the vehicle driving assist apparatus **10** continues applying the braking force to the own vehicle **100** in order to maintain the own vehicle moving speed V_{100} at the set vehicle moving speed V_{SET} . If the slope where the own vehicle **100** moves down is moderate, that is, if the own vehicle **100** moves down the slope having a small gradient, the vehicle driving assist apparatus **10** continues applying the relatively small braking force to the own vehicle **100** for a relatively long time.

(72) If the small braking force has been applied to the own vehicle **100** for a long time, a vibration such as a so-called self-excited vibration may occur due to thermal expansion of the brake pads and the brake discs. Such a vibration may lead to a discomfort of the driver DV of the own vehicle **100**.

(73) Accordingly, when the own vehicle **100** moves down the slope or the downward slope having a gradient smaller than a predetermined gradient threshold while the constant speed moving control is executed, the vehicle driving assist apparatus **10** pauses the constant speed moving control and executes the intermittent braking control. In particular, when an intermittent braking control start condition is satisfied, the vehicle driving assist apparatus **10** pauses the constant speed moving control and starts the intermittent braking control. The intermittent braking control start condition is a condition that the own vehicle **100** moves down the slope having a gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed.

(74) In particular, in this embodiment, the vehicle driving assist apparatus **10** applies the braking force to the own vehicle **100** while the constant speed moving control is executed. In this regard, when the braking force has been smaller than a predetermined value or a predetermined braking force threshold BK_TH for a predetermined time T_TH , the vehicle driving assist apparatus **10** determines that the own vehicle **100** moves down the slope having a gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed, and pauses the constant speed moving control and executes the intermittent braking control. In other words, when there is satisfied a condition that the braking force has been smaller than the predetermined braking force threshold BK_TH for the predetermined time T_TH while the constant speed moving control is executed, the vehicle driving assist apparatus **10** determines that the intermittent braking control start condition is satisfied and pauses the constant speed moving control and starts the intermittent braking control.

(75) In this embodiment, the predetermined braking force threshold BK_TH is set to a lower limit of the braking force which does not produce the self-excited vibration in the braking apparatus **22**.

(76) <Intermittent Braking Control>

(77) As shown in FIG. 4, when the braking force has been smaller than the predetermined braking force threshold BK_TH for the predetermined time T_TH while the constant speed moving control is executed, and the vehicle driving assist apparatus **10** starts the intermittent braking control (see a time t_{40}), the vehicle driving assist apparatus **10** controls the braking force applied to the own vehicle **100** or a vehicle applied braking force to zero once.

(78) Then, as shown in FIG. 4, when the own vehicle moving speed V_{100} increases to a fuel-cut process start vehicle moving speed threshold V_P1 (see a time t_{41}) while the intermittent braking control is executed, the vehicle driving assist apparatus **10** starts a fuel-cut process. In other words, when there is satisfied a fuel-cut process start condition that the own vehicle moving speed V_{100} increases to the fuel-cut process start vehicle moving speed threshold V_P1 while the intermittent braking control is executed, the vehicle driving assist apparatus **10** starts the fuel-cut process. The fuel-cut process start vehicle moving speed threshold V_P1 is set to a value greater than the set vehicle moving speed V_{SET} by a predetermined value or a first addition value dV_P1 .

(79) It should be noted that the fuel-cut process is a process to stop injecting fuel from fuel injectors of the internal combustion engine as the driving apparatus **21**. Thus, when the fuel-cut process is executed, a so-called engine brake is applied to the own vehicle **100** and as a result, the own vehicle **100** is decelerated. Further, when the accelerator pedal operation amount AP is zero, the vehicle driving assist apparatus **10** causes the fuel injectors to inject the fuel having an amount

necessary to operate the internal combustion engine. In this case, the engine brake is also applied to the own vehicle **100**, but the deceleration of the own vehicle **100** is smaller than the deceleration of the own vehicle **100** realized by the fuel-cut process. It should be noted that when the braking force is applied to the own vehicle **100**, the vehicle driving assist apparatus **10** also causes the fuel injectors to inject the fuel having an amount necessary to operate the internal combustion engine.

(80) Further, as shown in FIG. 4, when the own vehicle moving speed V_{100} increases to an intermittent braking process start vehicle moving speed threshold V_{P2} (see a time t_{42}) while the intermittent braking control is executed, the vehicle driving assist apparatus **10** starts an intermittent braking process to apply a predetermined braking force or a predetermined intermittent braking force BK_SET . In other words, when there is satisfied an intermittent braking process start condition that the own vehicle moving speed V_{100} increases to the intermittent braking process start vehicle moving speed threshold V_{P2} while the intermittent braking control is executed, the vehicle driving assist apparatus **10** starts the intermittent braking process. The predetermined intermittent braking force BK_SET is set to the braking force equal to or greater than the predetermined braking force threshold BK_TH . Further, the intermittent braking process start vehicle moving speed threshold V_{P2} is set to a value greater than the set vehicle moving speed V_SET by a predetermined value or a second addition value dV_{P2} . Furthermore, the second addition value dV_{P2} is set to a value greater than the first addition value dV_{P1} . It should be noted that at this time, the vehicle driving assist apparatus **10** continues the fuel-cut process.

(81) Further, as shown in FIG. 4, when the own vehicle moving speed V_{100} increases to a speed increasing control stop vehicle moving speed threshold V_{P3} or a predetermined upper limit vehicle moving speed (see a time t_{43}) while the intermittent braking control is executed, the vehicle driving assist apparatus **10** stops the intermittent braking control and starts the constant speed moving control. In other words, when there is satisfied a control stop condition that the own vehicle moving speed V_{100} increases to the speed increasing control stop vehicle moving speed threshold V_{P3} while the intermittent braking control is executed, the vehicle driving assist apparatus **10** stops the intermittent braking control and starts the constant speed moving control. The speed increasing control stop vehicle moving speed threshold V_{P3} is set to a value greater than the set vehicle moving speed V_SET by a predetermined value or a third addition value dV_{P3} . The third addition value dV_{P3} is set to a value greater than the second addition value dV_{P2} . It should be noted that at this time, the intermittent braking control is stopped and thus, the fuel-cut process and the intermittent braking process are stopped.

(82) On the other hand, as shown in FIG. 5, when the own vehicle moving speed V_{100} decreases to an intermittent braking process stop vehicle moving speed threshold V_{N1} (see a time t_{50}) while the intermittent braking control is executed, the vehicle driving assist apparatus **10** stops the intermittent braking process. In other words, when there is satisfied an intermittent braking process stop condition that the own vehicle moving speed V_{100} decreases to the intermittent braking process stop vehicle moving speed threshold V_{N1} while the intermittent braking control is executed, the vehicle driving assist apparatus **10** stops the intermittent braking process. The intermittent braking process stop vehicle moving speed threshold V_{N1} is set to a value smaller than the set vehicle moving speed V_SET by a predetermined value or a first subtraction value dV_{N1} . It should be noted that at this time, the vehicle driving assist apparatus **10** does not stop the fuel-cut process.

(83) Further, as shown in FIG. 5, when the own vehicle moving speed V_{100} decreases to a fuel-cut process stop vehicle moving speed threshold V_{N2} (see a time t_{51}) while the intermittent braking control is executed, the vehicle driving assist apparatus **10** stops the fuel-cut process. The fuel-cut process stop vehicle moving speed threshold V_{N2} is set to a value smaller than the set vehicle moving speed V_SET by a predetermined value or a second subtraction value dV_{N2} . The second subtraction value dV_{N2} is set to a value greater than the first subtraction value dV_{N1} .

(84) Further, as shown in FIG. 5, when the own vehicle moving speed V_{100} decreases to a speed

decreasing control stop vehicle moving speed threshold V_N3 (see a time $t52$) while the intermittent braking control is executed, the vehicle driving assist apparatus **10** stops the intermittent braking control and starts the constant speed moving control. In other words, when there is satisfied the control stop condition that the own vehicle moving speed $V100$ decreases to the speed decreasing control stop vehicle moving speed threshold V_N3 while the intermittent braking control is executed, the vehicle driving assist apparatus **10** stops the intermittent braking control and starts the constant speed moving control. The speed decreasing control stop vehicle moving speed threshold V_N3 is set to a value smaller than the set vehicle moving speed V_SET by a predetermined value or a third subtraction value dV_N3 . The third subtraction value dV_N3 is set to a value greater than the second subtraction value dV_N2 .

(85) When the vehicle driving assist apparatus **10** is configured to execute the intermittent braking control as described above, and the own vehicle **100** moves down the moderate slope, the braking force applied to the own vehicle **100** (i.e., the vehicle applied braking force) is controlled as shown in FIG. 6.

(86) When the vehicle applied braking force has been smaller than the predetermined braking force threshold BK_TH for the predetermined time T_TH (see a time $t60$) while the constant speed moving control is executed, the constant speed moving control is paused, and the intermittent braking control is started. When the intermittent braking control is started, the vehicle applied braking force is controlled to zero. Thus, the own vehicle moving speed $V100$ starts to increase.

(87) Thereafter, when the own vehicle moving speed $V100$ increases to the fuel-cut process start vehicle moving speed threshold V_P1 (see a time $t61$), the fuel-cut process is started. Thereafter, when the own vehicle moving speed $V100$ increases to the intermittent braking process start vehicle moving speed threshold V_P2 (see a time $t62$), the intermittent braking process is started. In this case, the braking force corresponding to the predetermined intermittent braking force BK_SET greater than the predetermined braking force threshold BK_TH is applied to the own vehicle **100**. Thereby, in an example shown in FIG. 6, the own vehicle moving speed $V100$ starts to decrease.

(88) When the own vehicle moving speed $V100$ decreases to the intermittent braking process stop vehicle moving speed threshold V_N1 (see a time $t63$), the intermittent braking process is stopped. Thereby, in the example shown in FIG. 6, the own vehicle moving speed $V100$ starts to increase. Thereafter, when the own vehicle moving speed $V100$ increases to the intermittent braking process start vehicle moving speed threshold V_P2 (see a time $t64$), the intermittent braking process is started. Thereby, in the example shown in FIG. 6, the own vehicle moving speed $V100$ starts to decrease. Thereafter, when the own vehicle moving speed $V100$ decreases to the intermittent braking process stop vehicle moving speed threshold V_N1 (see a time $t65$), the intermittent braking process is stopped. Thereby, in the example shown in FIG. 6, the own vehicle moving speed $V100$ starts to increase. Thereafter, when the own vehicle moving speed $V100$ increases to the intermittent braking process start vehicle moving speed threshold V_P2 (see a time $t66$), the intermittent braking process is started. Thereby, in the example shown in FIG. 6, the own vehicle moving speed $V100$ starts to decrease. Thereafter, when the own vehicle moving speed $V100$ decreases to the intermittent braking process stop vehicle moving speed threshold V_N1 (see a time $t67$), the intermittent braking process is stopped.

(89) <Advantages>

(90) As described above, when the own vehicle **100** moves down the moderate slope while the constant speed moving control is executed, the relatively small braking force may continue being applied to the own vehicle **100** for a relatively long time. If the small braking force continues being applied to the own vehicle **100** for a long time, the vibration such as a so-called self-excited vibration may occur due to thermal expansion of the brake pads and the brake discs. Such a vibration may lead to a discomfort of the driver DV of the own vehicle **100**.

(91) With the vehicle driving assist apparatus **10**, when the braking force has been smaller than the

predetermined braking force threshold BK_TH for the predetermined time T_TH while the constant speed moving control is executed, the intermittent control is executed. The braking force applied to the own vehicle **100** by the intermittent braking control is greater than the predetermined braking force threshold BK_TH. Thus, an occurrence of the self-excited vibration can be prevented.

(92) It should be noted that with the vehicle driving assist apparatus **10**, as described with reference to the example shown in FIG. **6**, applying and stopping applying the braking force to the own vehicle **100** are alternately carried out while the intermittent braking control is executed. Thus, the own vehicle moving speed V₁₀₀ can be maintained near the set vehicle moving speed V_SET while the intermittent braking control is executed.

(93) <Specific Operations of Vehicle Driving Assist Apparatus>

(94) Next, specific operations of the vehicle driving assist apparatus **10** will be described. The CPU of the ECU **90** of the vehicle driving assist apparatus **10** is configured or programmed to execute a routine shown in FIG. **7** with a predetermined calculation cycle. Thus, at a predetermined time, the CPU starts a process from a step **700** of the routine shown in FIG. **7** and proceeds with the process to a step **705** to determine whether the driving assist control is requested to be executed.

(95) When the CPU determines “Yes” at the step **705**, the CPU proceeds with the process to a step **710** to determine whether there is a preceding vehicle **200**.

(96) When the CPU determines “Yes” at the step **710**, the CPU proceeds with the process to a step **715** to execute a routine shown in FIG. **8**. Thus, when the CPU proceeds with the process to the step **715**, the CPU starts a process from a step **800** of the routine shown in FIG. **8** and proceeds with the process to a step **805** to determine whether the inter-vehicle distance D is greater than the target inter-vehicle distance D_TGT.

(97) When the CPU determines “Yes” at the step **805**, the CPU proceeds with the process to a step **810** to calculate the acceleration for controlling the inter-vehicle distance D to the target inter-vehicle distance D_TGT as the target acceleration GA_TGT, calculate the driving force to be applied to the own vehicle **100** in order to realize the acceleration corresponding to the target acceleration GA_TGT as a target driving force DR_TGT, and control the operations of the driving apparatus **21** so as to apply the driving force corresponding to the target driving force DR_TGT from the driving apparatus **21** to the own vehicle **100**. Next, the CPU proceeds with the process to a step **795** of the routine shown in FIG. **7** via a step **895** to terminate executing this routine once.

(98) On the other hand, when the CPU determines “No” at the step **805**, the CPU proceeds with the process to a step **815** to determine whether the inter-vehicle distance D is smaller than the target inter-vehicle distance D_TGT.

(99) When the CPU determines “Yes” at the step **815**, the CPU proceeds with the process to a step **820** to calculate the deceleration for controlling the inter-vehicle distance D to the target inter-vehicle distance D_TGT as the target deceleration GD_TGT, calculate the braking force to be applied to the own vehicle **100** in order to realize the deceleration corresponding to the target deceleration GD_TGT as a target braking force BK_TGT, and control the operations of the braking apparatus **22** and/or the driving apparatus **21** so as to apply the braking force corresponding to the target braking force BK_TGT from the braking apparatus **22** and/or the driving apparatus **21** to the own vehicle **100**. Next, the CPU proceeds with the process to the step **795** of the routine shown in FIG. **7** via the step **895** to terminate executing this routine once.

(100) On the other hand, when the CPU determines “No” at the step **815**, the CPU proceeds with the process to a step **825** to calculate the driving force necessary to maintain the current own vehicle moving speed V₁₀₀ as the target driving force DR_TGT and control the operations of the driving apparatus **21** so as to apply the driving force corresponding to the target driving force DR_TGT from the driving apparatus **21** to the own vehicle **100**. Next, the CPU proceeds with the process to the step **795** of the routine shown in FIG. **7** via the step **895** to terminate executing this routine once.

(101) Further, when the CPU determines “No” at the step **710** of the routine shown in FIG. 7, the CPU proceeds with the process to a step **720** to determine whether the intermittent braking control is executed.

(102) When the CPU determines “Yes” at the step **720**, the CPU proceeds with the process to a step **725** to determine whether the intermittent braking process stop condition is satisfied.

(103) When the CPU determines “Yes” at the step **725**, the CPU proceeds with the process to a step **730** to execute a routine shown in FIG. 9. Thus, when the CPU proceeds with the process to the step **730**, the CPU starts a process from a step **900** of the routine shown in FIG. 9 and proceeds with the process to a step **905** to determine whether the own vehicle moving speed **V100** is smaller than the set vehicle moving speed **V_SET**.

(104) When the CPU determines “Yes” at the step **905**, the CPU proceeds with the process to a step **910** to calculate the acceleration for controlling the own vehicle moving speed **V100** to the set vehicle moving speed **V_SET** as the target acceleration **GA_TGT**, calculate the driving force to be applied to the own vehicle **100** in order to realize the acceleration corresponding to the target acceleration **GA_TGT** as the target driving force **DR_TGT**, and control the operations of the driving apparatus **21** so as to apply the driving force corresponding to the target driving force **DR_TGT** from the driving apparatus **21** to the own vehicle **100**. Next, the CPU proceeds with the process to the step **795** of the routine shown in FIG. 7 via a step **995** to terminate executing this routine once.

(105) On the other hand, when the CPU determines “No” at the step **905**, the CPU proceeds with the process to a step **915** to determine whether the own vehicle moving speed **V100** is greater than the set vehicle moving speed **V_SET**.

(106) When the CPU determines “Yes” at the step **915**, the CPU proceeds with the process to a step **920** to calculate the deceleration for controlling the own vehicle moving speed **V100** to the set vehicle moving speed **V_SET** as the target deceleration **GD_TGT**, calculate the braking force to be applied to the own vehicle **100** in order to realize the deceleration corresponding to the target deceleration **GD_TGT** as the target braking force **BK_TGT**, and control the operations of the braking apparatus **22** and/or the driving apparatus **21** so as to apply the braking force corresponding to the target braking force **BK_TGT** from the braking apparatus **22** and/or the driving apparatus **21** to the own vehicle **100**. Next, the CPU proceeds with the process to the step **795** of the routine shown in FIG. 7 via the step **995** to terminate executing this routine once.

(107) On the other hand, when the CPU determines “No” at the step **915**, the CPU proceeds with the process to a step **925** to calculate the driving force necessary to maintain the current own vehicle moving speed **V100** as the target driving force **DR_TGT** and control the operations of the driving apparatus **21** so as to apply the driving force corresponding to the target driving force **DR_TGT** from the driving apparatus **21** to the own vehicle **100**. Next, the CPU proceeds with the process to the step **795** of the routine shown in FIG. 7 via the step **995** to terminate executing this routine once.

(108) Further, when the CPU determines “No” at the step **725** of the routine shown in FIG. 7, the CPU proceeds with the process to a step **740** to execute a routine shown in FIG. 10. Thus, when the CPU proceeds with the process to the step **740**, the CPU starts a process from a step **1000** of the routine shown in FIG. 10 and proceeds with the process to a step **1005** to determine whether the own vehicle moving speed **V100** increases.

(109) When the CPU determines “Yes” at the step **1005**, the CPU proceeds with the process to a step **1010** to determine whether the own vehicle moving speed **V100** is equal to or greater than the fuel-cut process start vehicle moving speed threshold **V_P1**.

(110) When the CPU determines “Yes” at the step **1010**, the CPU proceeds with the process to a step **1015** to execute the fuel-cut process. Next, the CPU proceeds with the process to a step **1020** to determine whether the own vehicle moving speed **V100** is equal to or greater than the intermittent braking process start vehicle moving speed threshold **V_P2**.

(111) When the CPU determines “Yes” at the step **1020**, the CPU proceeds with the process to a step **1025** to execute the intermittent braking process. That is, the braking force corresponding to the predetermined intermittent braking force BK_SET is applied to the own vehicle **100**. Next, the CPU proceeds with the process to the step **795** of the routine shown in FIG. 7 via a step **1095** to terminate executing this routine once.

(112) On the other hand, when the CPU determines “No” at the step **1010** or **1020**, the CPU proceeds with the process directly to the step **795** of the routine shown in FIG. 7 via the step **1095** to terminate executing this routine once.

(113) Further, when the CPU determines “No” at the step **1005**, the CPU proceeds with the process to a step **1030** to determine whether the own vehicle moving speed V100 decreases.

(114) When the CPU determines “Yes” at the step **1030**, the CPU proceeds with the process to a step **1035** to determine whether the own vehicle moving speed V100 is equal to or smaller than the intermittent braking process stop vehicle moving speed threshold V_N1.

(115) When the CPU determines “Yes” at the step **1030**, the CPU proceeds with the process to a step **1040** to stop the intermittent braking process. Next, the CPU proceeds with the process to a step **1045** to determine whether the own vehicle moving speed V100 is equal to or smaller than the fuel-cut process stop vehicle moving speed threshold V_N2.

(116) When the CPU determines “Yes” at the step **1045**, the CPU proceeds with the process to a step **1050** to stop the fuel-cut process. Next, the CPU proceeds with the process to the step **795** of the routine shown in FIG. 7 via the step **1095** to terminate executing this routine once.

(117) On the other hand, when the CPU determines “No” at the step **1035** or **1045**, the CPU proceeds with the process directly to the step **795** of the routine shown in FIG. 7 via the step **1095** to terminate executing this routine once.

(118) Further, when the CPU determines “No” at the step **720** of the routine shown in FIG. 7, the CPU proceeds with the process to a step **735** to determine whether the intermittent braking control start condition is satisfied.

(119) Further, when the CPU determines “Yes” at the step **735**, the CPU proceeds with the process to the step **740** to execute the routine shown in FIG. 10 as described above and then, proceeds with the process to the step **795** to terminate executing this routine once.

(120) On the other hand, when the CPU determines “No” at the step **735**, the CPU proceeds with the process to the step **745** to execute the routine shown in FIG. 9 as described above and then, proceeds with the process to the step **795** to terminate executing this routine once.

(121) Further, when the CPU determines “No” at the step **705**, the CPU proceeds with the process to a step **750** to execute the ordinary accelerating-and-decelerating control.

(122) The specific operations of the vehicle driving assist apparatus **10** have been described.

(123) It should be noted that the invention is not limited to the aforementioned embodiments, and various modifications can be employed within the scope of the invention.

Claims

1. A vehicle driving assist apparatus comprising an electronic control unit which executes a constant speed moving control to autonomously control a moving speed of an own vehicle to a set moving speed, wherein the electronic control unit is configured to: while the constant speed moving control is being executed, determine whether the own vehicle moves down a slope having a gradient smaller than a predetermined gradient threshold when a braking force applied to the own vehicle by the constant speed moving control has been smaller than a predetermined braking force threshold for a predetermined time; based upon the determination that the own vehicle moves down the slope having the gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed even when the moving speed of the own vehicle is maintained to the set moving speed, pause the constant speed moving control and start an intermittent braking

control; and based upon the determination that the own vehicle does not move down the slope having the gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed, continue the constant speed moving control, wherein the intermittent braking control is a control to alternately apply and stop applying a braking force to the own vehicle by: applying the braking force when the moving speed of the own vehicle increases to an upper threshold greater than the set moving speed while stopping applying the braking force to the own vehicle; and stopping applying the braking force when the moving speed of the own vehicle decreases to a lower threshold smaller than the set moving speed while applying the braking force to the own vehicle, and wherein the braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control.

2. A vehicle comprising the vehicle driving assist apparatus as set forth in claim 1.

3. The vehicle driving assist apparatus as set forth in claim 1, wherein the electronic control unit is configured to pause the intermittent braking control and start the constant speed moving control when the moving speed of the own vehicle becomes equal to or greater than a predetermined upper limit vehicle moving speed greater than the upper threshold, or becomes equal to or smaller than a predetermined lower limit vehicle moving speed smaller than the lower threshold while the intermittent braking control is executed.

4. A non-transitory computer readable medium comprising a vehicle driving assist program that, when executed by a processor, performs a constant speed moving control to autonomously control a moving speed of an own vehicle to a set moving speed, wherein the vehicle driving assist program is configured to: while the constant speed moving control is being executed, determine whether the own vehicle moves down a slope having a gradient smaller than a predetermined gradient threshold when a braking force applied to the own vehicle by the constant speed moving control has been smaller than a predetermined braking force threshold for a predetermined time; based upon the determination that the own vehicle moves down the slope having the gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed even when the moving speed of the own vehicle is maintained to the set moving speed, pause the constant speed moving control and start an intermittent braking control; and based upon the determination that the own vehicle does not move down the slope having the gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed, continue the constant speed moving control, wherein the intermittent braking control is a control to alternately apply and stop applying a braking force to the own vehicle by: applying the braking force when the moving speed of the own vehicle increases to an upper threshold greater than the set moving speed while stopping applying the braking force to the own vehicle; and stopping applying the braking force when the moving speed of the own vehicle decreases to a lower threshold smaller than the set moving speed while applying the braking force to the own vehicle, and wherein the braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control.

5. A vehicle driving assist method of executing a constant speed moving control to autonomously control a moving speed of an own vehicle to a set moving speed, wherein the vehicle driving assist method comprises steps of: while the constant speed moving control is being executed, determining whether the own vehicle moves down a slope having a gradient smaller than a predetermined gradient threshold when a braking force applied to the own vehicle by the constant speed moving control has been smaller than a predetermined braking force threshold for a predetermined time; based upon the determination that the own vehicle moves down the slope having the gradient smaller than the predetermined gradient threshold while the constant speed moving control is executed even when the moving speed of the own vehicle is maintained to the set moving speed, pausing the constant speed moving control and starting an intermittent braking control; and based upon the determination that the own vehicle does not move down the slope having the gradient

smaller than the predetermined gradient threshold while the constant speed moving control is executed, continuing the constant speed moving control, wherein the intermittent braking control is a control to alternately apply and stop applying a braking force to the own vehicle by: applying the braking force when the moving speed of the own vehicle increases to an upper threshold greater than the set moving speed while stopping applying the braking force to the own vehicle; and stopping applying the braking force when the moving speed of the own vehicle decreases to a lower threshold smaller than the set moving speed while applying the braking force to the own vehicle, and wherein the braking force applied to the own vehicle by the intermittent braking control is greater than the braking force applied to the own vehicle by the constant speed moving control.
